Hospital Booking System Specification

Purpose

The purpose of this Hospital Booking System is to provide a convenient way for patients to book appointments, view doctor information and availability, check token numbers and queue status, and cancel or reschedule appointments online. It also helps administrators manage doctors, schedules, appointments, and patient queues efficiently, reducing manual work and improving the overall appointment booking process.

Functional Requirements Specification

FR-1: View Doctor Information and Availability

1.1 Description:

The system shall allow patients to view doctors' information, specializations, and available appointment dates and times.

1.2 Actor:

Patient

1.3 Precondition:

- The patient can access the doctor information page.
- Doctor information and schedules are available in the system.

1.4 Main Flow:

1. The patient opens the doctor list.
2. The system displays the available doctors and their information.
3. The patient selects a doctor.
4. The system displays the selected doctor's details and specialization.
5. The system displays the available appointment dates and time slots.
6. The patient reviews the available schedules.

1.5 Alternative Flow:

- If no doctors are available, the system displays an appropriate message.
- If the selected doctor has no available appointment slots, the system informs the patient.
- The system prevents the patient from selecting unavailable time slots.

1.6 Expected Result:

The patient can view doctor information and identify available dates and times for an appointment.

FR-2: Book an Appointment, submit an optional visit Reason, and Reason, and Receive a Token

2.1 Description:

The system shall allow patients to book an book appointments online by selecting a doctor,

an available date, and an available time slot. Patient can optionally submit a reason for their visit . The system shall generate a token number after a successful booking.

2.2 Actor:

Patient

2.3 Precondition:

- The patient is logged in.
- The selected doctor has available appointment slots.
- The patient can access the appointment booking function.

2.4 Main Flow:

1. The patient opens the appointment booking page.
2. The patient selects a doctor.
3. The system displays the doctor’s available dates and times.
4. The patient selects an available date and time.
5. The patient optionally enters a reason for the visit.
6. The system displays the selected appointment details.
7. The patient confirms the booking.
8. The system verifies that the selected time slot is still available.
9. The system creates and saves the appointment.
10. The system generates a token number for the appointment.
11. The system displays the booking confirmation and token number.

2.5 Alternative Flow:

- If the selected time slot is no longer available, the system prevents the booking and asks the patient to select another available slot.
- If the patient does not enter a visit reason ,the system continues the booking process.
- If the booking cannot be saved, the system displays an appropriate error message.

2.6 Expected Result:

The appointment is successfully recorded, and the patient receives a token number associated with the appointment.

FR-3: View Personal Token and Appointment Status

3.1 Description:

The system shall allow patients to view their personal token number, the current queue position, and their appointment status.

3.2 Actor:

Patient

3.3 Precondition:

- The patient is logged in.
- The patient has a valid appointment and an associated token number.

3.4 Main Flow:

1. The patient opens the appointment status or queue status page.
2. The system retrieves the patient ‘s appointment and token information.
3. The system retrieves the current queue information.
4. The system displays the patient’s personal token number.
5. The system displays the current token being served,
6. The system displays the patient’s personal token position.
7. The system displays the appointment or token status.
8. The system updates the displayed information when the queue status changes.

3.5 Alternative Flow:

- If the patient has no appointment or token, the system displays an appropriate message.
- If the queue information cannot be retrieved, the system informs the patient that the information is temporarily unavailable.

3.6 Expected Result:

The patient can view their token number, current queue information, and appointment status.

FR-4: Cancel or Reschedule an Appointment/Arrange a follow-up

4.1 Description:

The system shall allow patients to cancel or reschedule their upcoming appointments and arrange follow-up appointments, subject to appointment status and available schedules.

4.2 Actor:

Patient

4.3 Precondition:

- The patient is logged in.
- The patient has an upcoming appointment.
- The appointment belongs to the logged-in patient.

4.4 .1 Main Flow — Cancel Appointment:

1. The patient opens the appointment details page.
2. The system displays the patient’s upcoming appointments.
3. The patient selects an appointment to cancel.
4. The system displays the selected appointment details.
5. The patient confirms the cancellation.
6. The system changes the appointment status to Cancelled.
7. The system updates the appointment and queue records as applicable.
8. The system displays a cancellation confirmation.

4.4.2 Main Flow — Reschedule Appointment:

1. The patient selects an upcoming appointment.
2. The patient chooses the reschedule option.

3. The system displays available alternative dates and time slots.
4. The patient selects a new available date and time.
5. The system verifies the availability of the selected slot.
6. The system updates the appointment with the new date and time.
7. The system displays the updated appointment details.

4.4.3 Main Flow —Arrange a follow up:

1. The patient open the appointment details page.
2. The patient select the follow-up appointment option.
3. The system displays available doctors, dates, and time slots.
4. The patient selects suitable date and time.
5. The system verifies the selected time slot.
6. The system saves the follow-up appointment.
7. The system displays the follow-up appointment confirmation.

4.6 Alternative Flow:

- If the selected appointment is not eligible for cancellation or rescheduling, the system informs the patient.
- If the selected reschedule slot is unavailable, the system asks the patient to choose another slot.
- If the follow-up appointment slots are available, the system informs the patient.
- If the appointment update fails, the system displays an error message.

4.7 Expected Result:

The appointment is successfully cancelled , rescheduled or arranged as a follow-up appointment. The system updates the appointment records and queue information as applicable.

FR-5: Manage Doctors Information and Schedules

5.1 Description:

The system shall allow administrators to add, edit and remove (or) manage doctor information, doctor schedules, and appointments availability.

5.2 Actor:

Admin

5.3 Precondition:

The administrator is logged in.

The administrator has permission to manage doctor information and schedules.

5.4 Main Flow :

1. The admin opens the management dashboard.
2. The admin selects the doctor management function.
3. The system displays the list of doctors.

4. The admin adds, edits, or removes doctor information.
5. The admin opens the schedule management page.
6. The system displays existing doctor schedules.
7. The admin creates or updates doctor schedules.
8. The system validates the submitted information.
9. The system saves the changes.
10. The system updates doctor information and availability in the patient interface.

5.5 Alternative Flow:

- If the admin submits invalid information, the system displays an error message.
- If the selected doctor does not exist, the system displays an appropriate message.
- If a schedule cannot be updated , the system informs the admin.

5.6 Expected Result:

The administrator can manage doctor information, schedules , and updated information is reflected in the patient interface.

FR-6: View Dashboard , Appointment, and Token Records (read only)

6.1 Description:

The system shall allow administrators to view the dashboard , appointment records, and token queue information in read-only mode without modifying or deleting records .

6.2 Actor:

Admin

6.3 Precondition:

The administrator is logged in.

Appointment and token records are available in the system.

6.4 Main Flow :

1. The admin opens the management dashboard.
2. The system displays the dashboard information.
3. The admin selects the appointment records section.
4. The system displays the list of patient appointments.
5. The admin selects an appointment to view its details.
6. The system displays the appointment information and current status.
7. The admin opens the token records section.
8. The system displays token numbers and queue information.
9. The admin reviews the displayed records.

6.5 Alternative Flow:

- If no appointment records are available, the system displays an appointment message.
- If token records cannot be retrieved , the system informs the admin.
- The system prevents unauthorized modifications to read-only records.

6.6 Expected Result:

The administrator can manage doctor information, schedules , and updated information is reflected in the patient interface.

- If the selected doctor does not exist, the system displays an appropriate message.
- If a schedule cannot be updated , the system informs the admin.

6.6 Expected Result:

The administrator can manage doctor information, schedules , and updated information is reflected in the patient interface.

FR-7: Manage Patient Account Access (Ban/Restore)

7.1 Description:

The system shall allow administrators to manage patient accounts access by banning or restoring patient accounts when necessary.

7.2 Actor:

Admin

7.3 Precondition:

- The administrator is logged in.
- The administrator has permission to manage patient accounts.
- Patient account records are available in the system.

7.4 Main Flow :

1. The admin opens the patient account management page.
2. The system displays the list of registered patient accounts.
3. The admin selects a patient account.
4. The system displays the selected patient’s account information and status.
5. The admin chooses to ban or restore the patient account,.
6. The system updates the patient account status.
7. The system saves the changes.
8. The system displays a confirmation message.

7.5 Alternative Flow:

- If the selected patient account does not exist, the system displays an appropriate error message.
- If the admin does not have permission to manage the account, the system denies the action.
- If the account status cannot be updated , the system displays an error message .

7.6 Expected Result:

The administrator can ban or restore patient accounts and the system updates the account access status accordingly.

FR-8: Register, Sign In, and Manage User Profile

8.1 Description:

The system shall allow patients and administrators to sign in using their registered credentials. Patients can register new accounts and manage their personal profile information, while administrators can access and manage their account information.

8.2 Actor:

Admin/Patient

8.3 Precondition:

- The patient or administrator can access the system.
- The patient or administrator has valid login credentials.
- The patient has not logged in yet for registration.
- The user has an account to access profile management.

8.4 Main Flow :

1. The patient or admin opens the registration or login page.
2. For registration, the patient enters the required personal information.
3. The system validates the submitted information.
4. The system creates a new patient account if the information is valid.
5. The patient or admin enters their registered login credentials.
6. The system verifies the login credentials.
7. The system grants access according to the user's role.
8. The patient or admin opens the profile management page.
9. The system displays the user's current profile information.
10. The user updates the necessary profile information.
11. The system validates and saves the updated information.
12. The system displays a profile update confirmation.

8.5 Alternative Flow:

- If the registration information is invalid, the system displays an error message and requests correction.
- If the login credentials are incorrect, the system displays an error message.
- If the profile information is invalid, the system requests the user to correct it.
- If the profile update fails, the system displays an appropriate error message.

8.6 Expected Result:

Patients and administrators can successfully sign in and manage their profile information. Patients can also register new accounts, and the system provides access according to the user's role.

FR-9: Receive and Manage In-App Notifications

9.1 Description:

The system shall allow patients and administrators to receive and manage in-app notifications related to appointments, booking confirmations, appointment status changes,

and important system updates.

9.2 Actor:

Admin/Patient

9.3 Precondition:

- The patient or administrator has logged in.
- The system has notification information available.
- The user can access the notification function.

9.4 Main Flow :

1. The patient or admin opens the notification page.
2. The system retrieves the user’s notification records.
3. The system displays the available notifications.
4. The user selects a notification to view its details.
5. The system displays the selected notification information.
6. The user marks the notification as read.
7. The system updates the notification status.
8. The user can manage their notification list.
9. The system saves the updated notification information.

9.5 Alternative Flow:

- If there are no notifications, the system displays an appropriate message.
- If notification information cannot be retrieved, the system informs the user that the information is temporarily unavailable.
- If the notification status cannot be updated, the system displays an error message..

9.6 Expected Result:

Patients and administrators can receive, view, and manage in-app notifications, and the system updates notification statuses accordingly.